All mixed identities are singular in groups with no algebraicity

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joint work with Michael Pinsker

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Outline

1. Background
   - Equations in groups
   - Mixed identities
2. On algebraic closure
3. The result
4. Strategy
5. A topological extra
6. Bibliografy

Words with constants

$G :=$ a group.

Definition (Word with constants)

A **word with constants** for G in r many variables is

$$w(x_1, \ldots, x_r) := \gamma_n \; x_{\iota_n}^{\epsilon_n} \; \gamma_{n-1} \; x_{\iota_{n-1}}^{\epsilon_{n-1}} \; \cdots \; \gamma_1 \; x_{\iota_1}^{\epsilon_1} \; \gamma_0$$

for $\gamma_0, \ldots, \gamma_n \in G$, $\epsilon_1, \ldots, \epsilon_n \in \{-1, 1\}$, and $\iota_1, \ldots, \iota_n \in \{1, \ldots, r\}$.

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I.e., $w \in G * F_r$.

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I.e., a **term** in $\mathcal{L} := \{\cdot, ^{-1}, (\gamma)_{\gamma \in G}\}$.

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w is **singular** if "forgetting constants" and reducing we get 1:

$$y^{-1} x \gamma_2 x^{-1} \gamma_1 y \xRightarrow[\text{constants}]{\text{forget}} y^{-1} x x^{-1} y \mapsto y^{-1} y \mapsto 1 \, .$$

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Otherwise, w is **regular** (a.k.a. non-singular).

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- when can we **solve** w?
  $\exists g_1, \ldots, g_r \in G$ s.t. $w(g_1, \ldots, g_r) = 1$;
- when is w a **mixed identity**?
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These notions are in tension!
$\forall \overline{g} \in G \ w(\overline{g}) = 1 \ \Rightarrow$ for $\gamma \in G \setminus \{1\}$, $\nexists \overline{g} \in G \, \gamma w(\overline{g}) = 1$.

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Solving equations

Conjecture (Klyachko and Thom 2017)

G a group, $w(x_1, \ldots, x_r)$ regular.
Then, there is $H \supseteq G$ s.t. $\exists h_1, \ldots, h_r \in H \ w(h_1, \ldots, h_r) = 1$.

- $r = 1$ this is the Kervaire-Laudenbach Conjecture:
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  - OpenAI 2026b 🤖 (presumably?): true for all groups.*

*OpenAI 2026a implies a weaker version of the Klyachko-Thom conjecture.

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Theorem (Adeleke and Holland 1994)

$G := \mathrm{Aut}(\mathbb{Q}; <)$. w with no constants and non-trivial.
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$G := \mathrm{Aut}(\mathbb{Q}; <)$. w with no constants and non-trivial.
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Theorem (Mycielski 1987; Lyndon 1990)

$G := \mathrm{Sym}(\mathbb{N})$. w with no constants, non-trivial, and $w \neq x^n$ $(n \geq 2)$.
Then, $w(G^r) = G$.

An interlude on oligomorphicity

$\mathrm{Aut}(\mathbb{Q};<)$ and $\mathrm{Sym}(\mathbb{N})$ are both oligomorphic:

Definition (Oligomorphic)

$|\Omega| = \aleph_0$. $G \curvearrowright \Omega$ is **oligomorphic** if for all $n \in \mathbb{N}$, $G \curvearrowright \Omega^n$ has finitely many orbits in the action $g \cdot (a_1, \ldots, a_n) \mapsto (ga_1, \ldots, ga_n)$.

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Ryll-Nardzweski: $G = \mathrm{Aut}(M)$ oligomorphic $\Leftrightarrow$ M is ω-categorical.*

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Algebraic examples: $\mathrm{GL}(\aleph_0, \mathbb{F}_q)$, $\mathrm{Sp}(\aleph_0, \mathbb{F}_q)$, $\mathrm{U}(\aleph_0, \mathbb{F}_q)$, $\mathrm{O}(\aleph_0, \mathbb{F}_q)$.

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Fraïssé:* hereditary class $\mathcal{C}$ with the **amalgamation property**:

(AP) for $A, B_0, B_1 \in \mathcal{C}$, and embeddings $f_i : A \to B_i$ $(i \in \{0, 1\})$, there is $C \in \mathcal{C}$ and embeddings $g_i : B_i \to C$ $(i \in \{0, 1\})$ s.t. $g_1 \circ f_1 = g_2 \circ f_2$.

*Assuming language is countable and purely relational.

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Often, we specify homogeneous structures by their age:
random graph, generic triangle-free graph, random poset...

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Mixed identities

Definition (MIF and lawless)

$w(x_1, \ldots, x_r)$ is **a mixed identity** if $w(G^r) = 1$.
A mixed identity without constants is a **law**.
G is **mixed-identity-free (MIF)**: no (non-trivial) mixed identities.
G is **lawless**: no (non-trivial) laws.

- Mixed identities describe the structure of a group/elements:

$$\forall x \; x^2 = 1 \quad \text{or} \quad \forall x \; x\gamma x^{-1}\gamma^{-1} = 1$$

- we would expect "rich group"$\Rightarrow$ "few" mixed identities.

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Theorem (Macpherson 1986)

$G \curvearrowright \Omega$ oligomorphic $\Rightarrow$ G is lawless.

Mixed identities in the oligomorphic/homogeneous context

Several automorphism groups of homogeneous structures are MIF:

- random graph, generic triangle free graph, Urysohn space (Etedadialiabadi, Gao, Le Maître, and Melleray 2021);
- transitive with free amalgamation (Ghadernezhad and de la Nuez González 2019; Bodirsky, Schneider, and Thom 2025);
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- $\mathrm{Sym}(\mathbb{N})$: $[x,t]^6$ where t is a transposition
  (Hull and Osin 2016);
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- random poset, random permutation
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Sometimes we have mixed identities!

- $\mathrm{Sym}(\mathbb{N})$: $[x, t]^6$ where t is a transposition
  (Hull and Osin 2016);
- $\mathrm{GL}(\aleph_0, \mathbb{F}_q)$
  (Bradford, Schneider, and Thom 2023);
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Mixed identities in the oligomorphic/homogeneous context

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In the first two cases, we know all mixed identities are singular!

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Conjecture (Bodirsky, Schneider, and Thom 2025)

$G \curvearrowright \Omega$ oligomorphic $\Rightarrow$ all mixed identities of G are singular.

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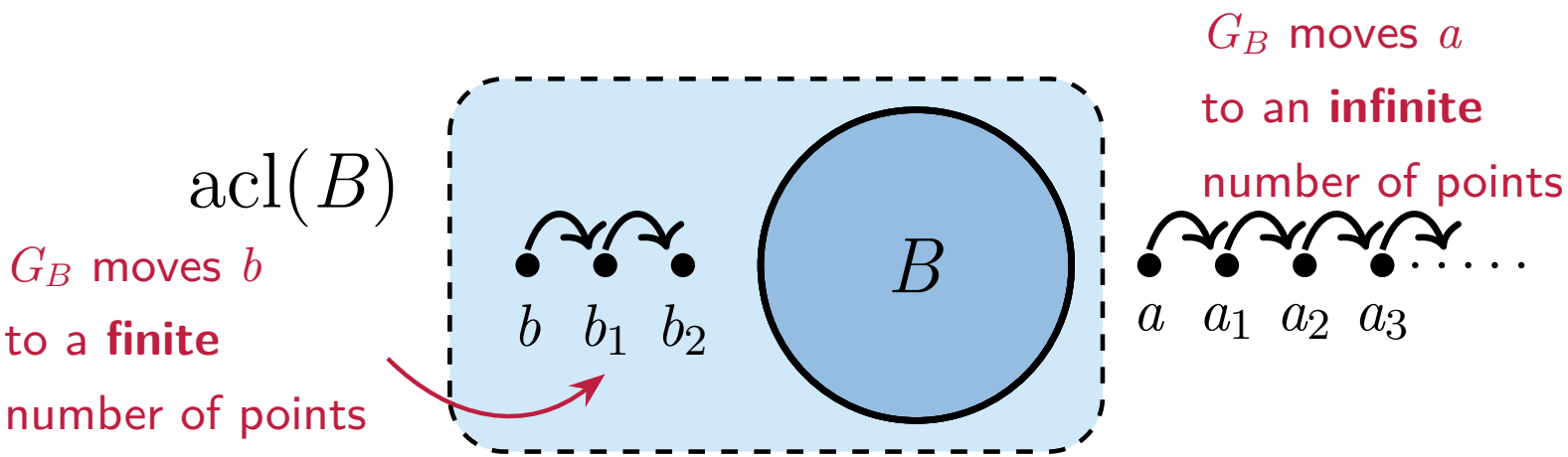
Which general conditions imply that all mixed identities are singular?

Algebraic closure

Definition (algebraic closure)

$G \curvearrowright \Omega$. $B \subseteq \Omega$ finite.

$$\mathrm{acl}(B) := \{a \in \Omega \mid a \text{ has finite } G_B\text{-orbit}\} .$$



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Lemma (Π. M. Neumann 1976)

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Theorem (Bardyla, Elliott, Mitchell, and Péresse 2025)

$G \curvearrowright \Omega$ with no algebraicity. Then, for distinct semigroup words w, v in one variable, there is $g \in G$, s.t. $w(g) \neq v(g)$.

Our result

Theorem (MP 2026)

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- automorphism groups of homogeneous structures whose age has the **strong amalgamation property**:
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  - confirms the Bodirsky, Schneider, and Thom 2025 conjecture for several cases beyond previous techniques:
    $(\mathbb{Q}; <)$ and reducts, 2-homogeneous semilinear orders of negative type, homogeneous local order $S(2)$, homogeneous C relations;

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- Weakly branch groups (include Grigorchuk's group).
  Has mixed identities (Jacobson 2021; Le Boudec and Matte Bon 2022);
- Any wreath product $G \wr H$, or direct product $G \times H$ where $G \curvearrowright \Omega$ has no algebraicity.
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$$\{(g_1, \dots, g_r) \in G^r \mid w(g_1, \dots, g_r) \neq 1\}$$
is dense in G^r wrt the topology of pointwise convergence.[†]
- We do not need the action to be faithful.

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Proof strategy[‡]

For $g \in G$, $a_0 \in \Omega$, the images of a_0 with respect to subwords of w create a sequence $(a_0, a_1, a_1', \ldots, a_n, a_n', a_{n+1})$ with

$$g \cdot (a_1, \ldots, a_n) = (a_1', \ldots, a_n') \ .$$

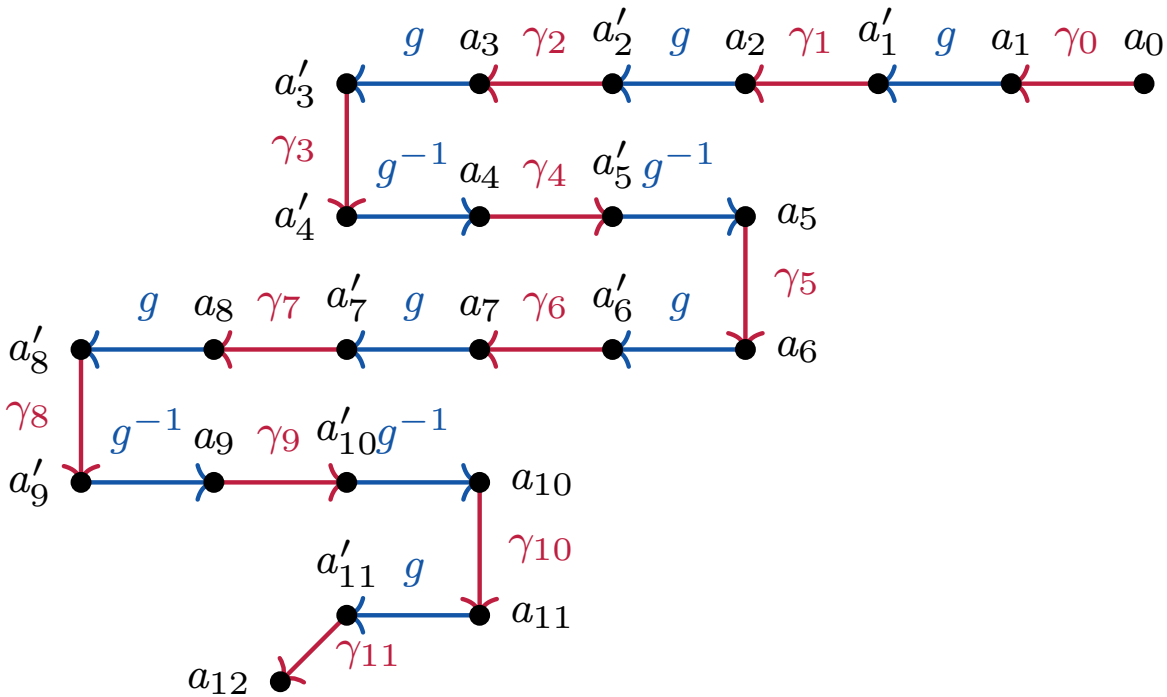
[‡] w with one variable.

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$$w(x) = \gamma_{11} x \gamma_{10} x^{-1} \gamma_9 x^{-1} \gamma_8 x \gamma_7 x \gamma_6 x \gamma_5 x^{-1} \gamma_4 x^{-1} \gamma_3 x \gamma_2 x \gamma_1 x \gamma_0$$

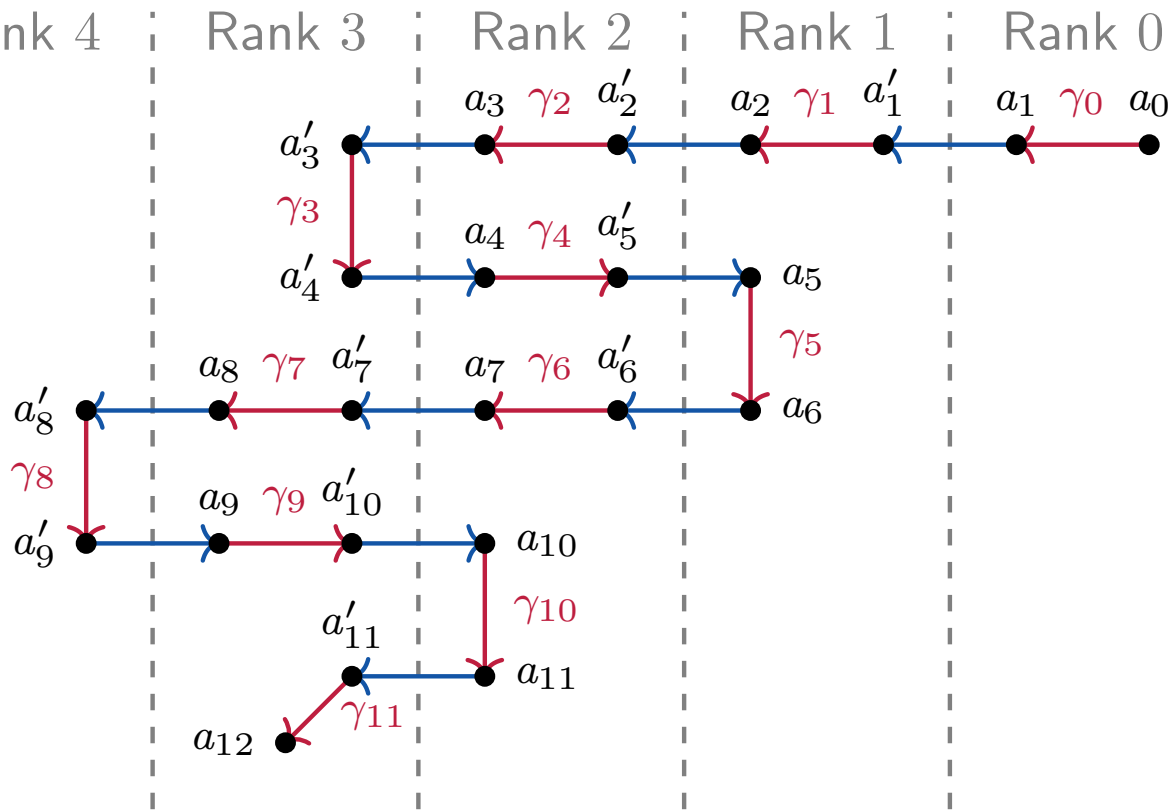


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Rank elements according to their position in $\mathbb{Z}$.

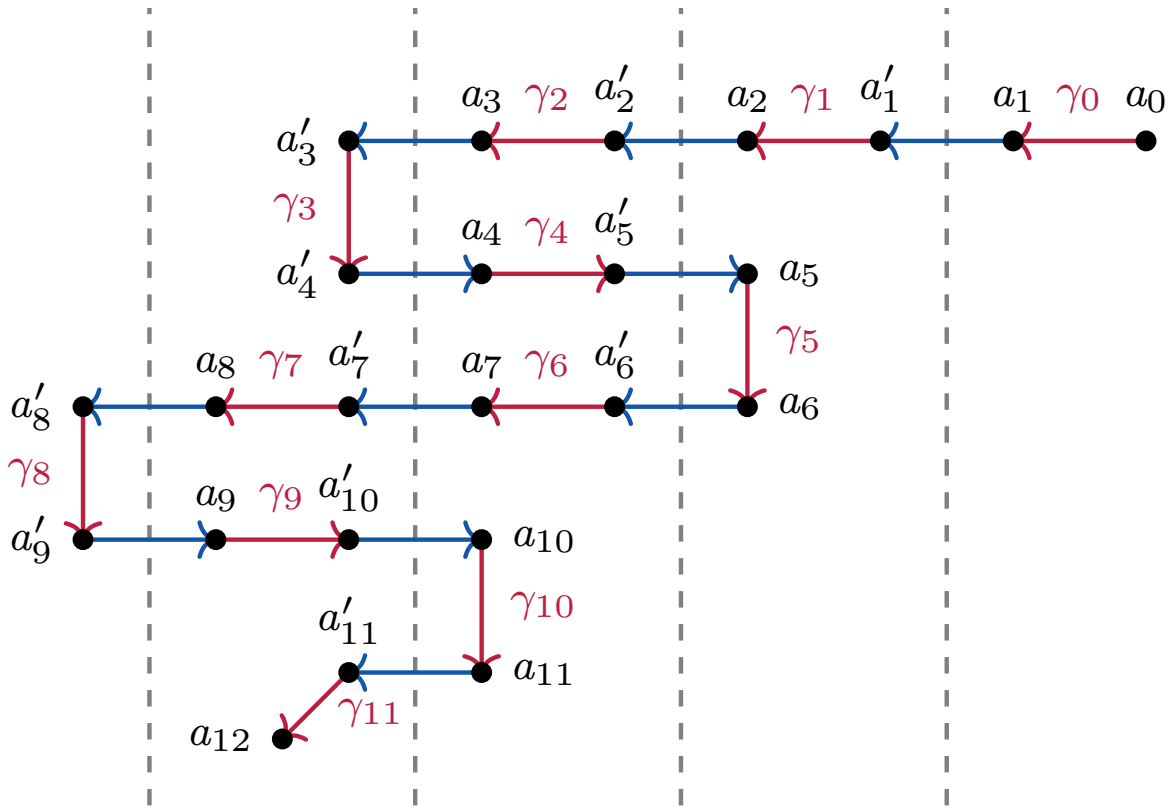


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We want g so that elements of different ranks are independent!

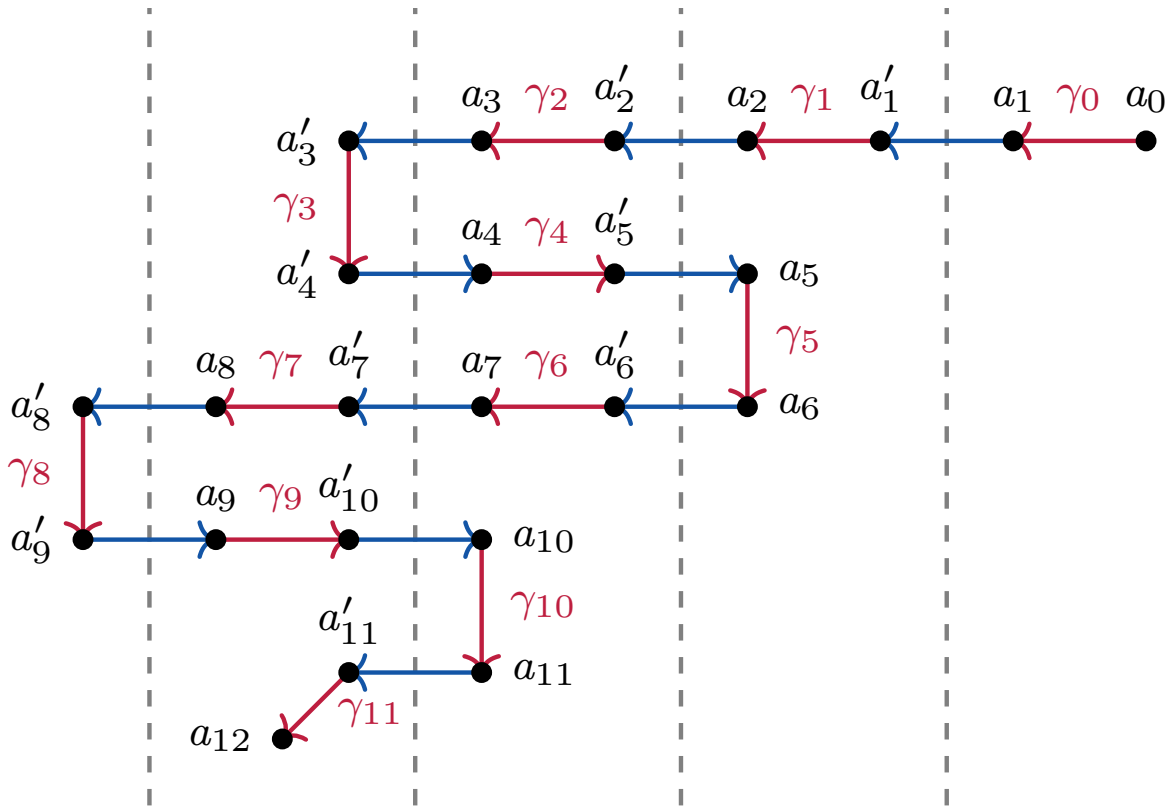


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We cannot guarantee elements of same rank to be distinct!

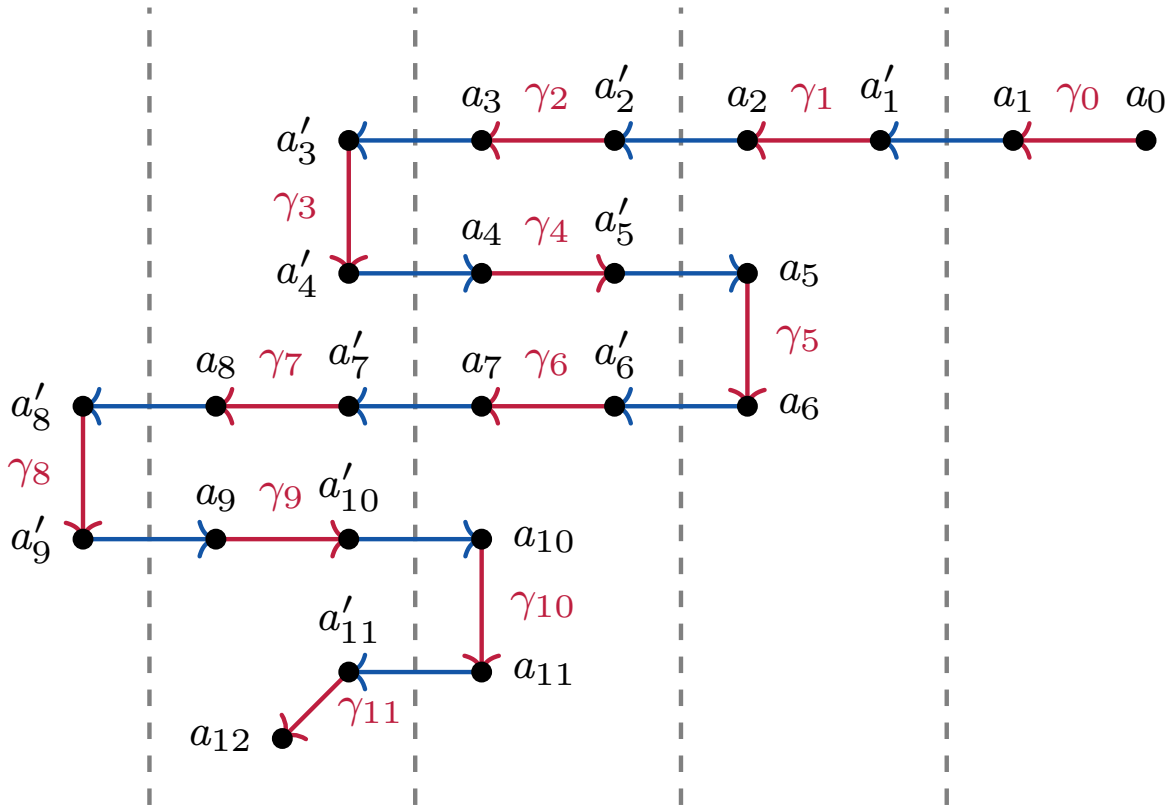


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Use Neumann's Lemma to guarantee independence (looking ahead!)

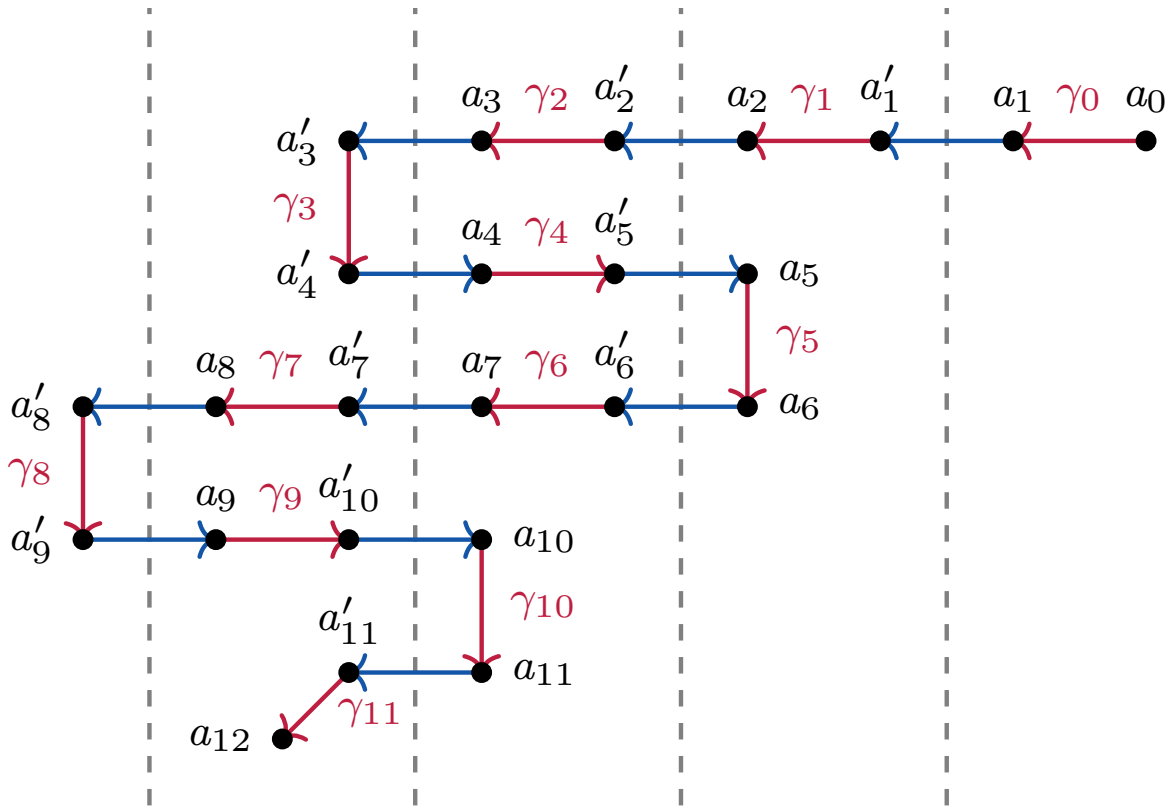


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Induction proof: challenge is balancing inductive hypothesis!



Geometric properties underlying the proof

With algebraicity, we need further geometric properties and a strengthening of Neumann's Lemma.

Definition (pregeometry)

(Ω, cl) for $\mathrm{cl} : \mathcal{P}(\Omega) \to \mathcal{P}(\Omega)$ is a **pregeometry** if it satisfies:

(REFLEXIVITY)	$A \subseteq \mathrm{cl}(A)$;
(TRANSITIVITY)	$\mathrm{cl}(\mathrm{cl}(A)) = \mathrm{cl}(A)$;
(MONOTONICITY)	if $A \subseteq B$, then $\mathrm{cl}(A) \subseteq \mathrm{cl}(B)$;
(FINITE CHARACTER)	$\mathrm{cl}(A) = \bigcup\{\mathrm{cl}(C) \mid C \subseteq A \text{ finite}\}$
(EXCHANGE)	if $a \in \mathrm{cl}(A \cup \{b\}) \setminus \mathrm{cl}(A)$, then $b \in \mathrm{cl}(A \cup \{a\})$.

It is **G-invariant** if for $g \in G$, $\mathrm{cl}(gA) = g\,\mathrm{cl}(A)$.

(Ω, cl) is **modular** if for all closed $A, B \subseteq \Omega$,

$$\dim(A \cup B) = \dim(A) + \dim(B) - \dim(A \cap B)\ .$$

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Higher Neumann pregeometries

Definition

Let $G \curvearrowright \Omega$. G-invariant (Ω, cl) is **higher Neumann** if for each $B, C \subseteq \Omega$ finite, $a \in \Omega$ such that $\mathrm{cl}(a) \cap \mathrm{cl}(B) = \mathrm{cl}(\emptyset)$, and $\Sigma \subseteq G$ finite, there is $g \in G_B$ such that

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Theorem (MP 2026)

Let $G \curvearrowright \Omega$. Let (Ω, cl) be a G-invariant infinite-dimensional higher Neumann modular pregeometry.
Then all mixed identities of G are singular.

Higher Neumann pregeometries

Definition

Let $G \curvearrowright \Omega$. G-invariant (Ω, cl) is **higher Neumann** if for each $B, C \subseteq \Omega$ finite, $a \in \Omega$ such that $\mathrm{cl}(a) \cap \mathrm{cl}(B) = \mathrm{cl}(\emptyset)$, and $\Sigma \subseteq G$ finite, there is $g \in G_B$ such that

$$\mathrm{cl}\left(\bigcup_{\gamma \in \Sigma} \gamma g(a)\right) \cap \mathrm{cl}(C) = \mathrm{cl}(\emptyset)\ .$$

Theorem (MP 2026)

Let $G \curvearrowright \Omega$. Let (Ω, cl) be a G-invariant infinite-dimensional higher Neumann modular pregeometry.
Then all mixed identities of G are singular.

Applies to $\mathrm{GL}(\kappa, K)$ and $\mathrm{PGL}(\kappa, K)$ for $\kappa \geq \aleph_0$, recovering results of Bradford, Schneider, and Thom 2023.

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Bradford, Ersoy, Schneider, and Thom 2026:
also yields the result for $\mathrm{Sp}(\aleph_0, \mathbb{F}_q)$, $\mathrm{SU}(\aleph_0, \mathbb{F}_q)$, $\mathrm{O}(\aleph_0, \mathbb{F}_q)$.

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▸ More on the proof

Microsupported actions

Several of the examples coming from topology are microsupported:

Definition

$\mathbf{\Omega} = (\Omega; \tau)$ Hausdorff, $G \leq \mathrm{Homeo}(\mathbf{\Omega})$. G is **microsupported** if for every non-empty open set $U \subseteq \Omega$, $G_{\Omega \setminus U} \neq \{1\}$.

Theorem (Nekrashevych 2022)

If G is microsupported, then G is lawless.

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If G is microsupported, then all of its mixed identities are singular.

Different proof idea: lifting lawlessness through microsupport.

▸ See proof

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Thank you for your time!

Recap:

- All mixed identities being singular is a strong form of lawlessness;
- We prove this for:
  - groups with no algebraicity;
  - groups with microsupported actions.
- Key new idea: use global geometric properties of actions instead of properties of constants (controlling products of constants).

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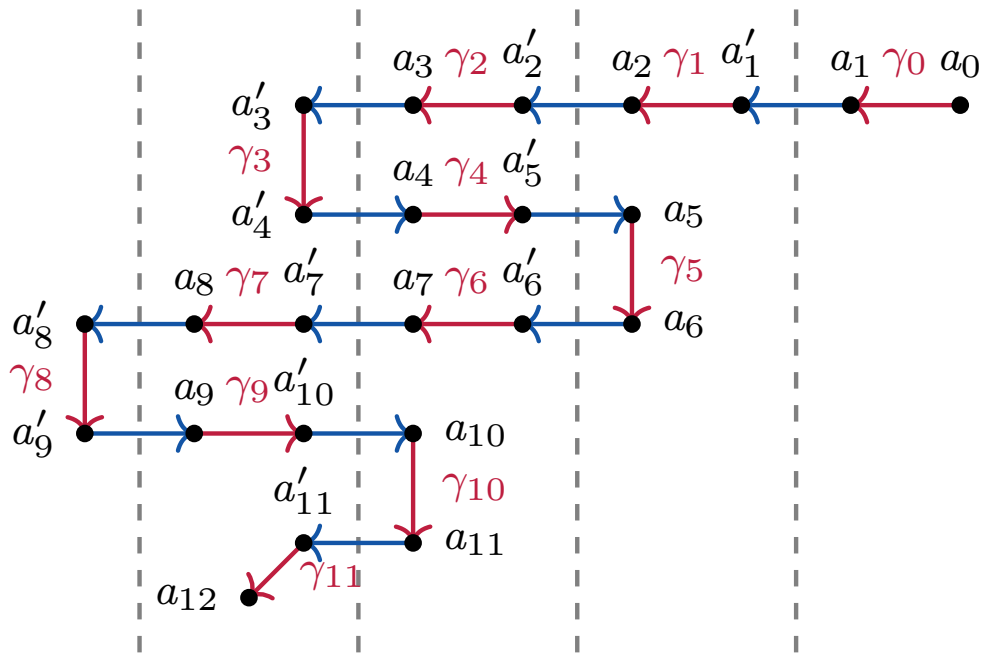
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Why these geometric conditions?

Induction splits into two cases:

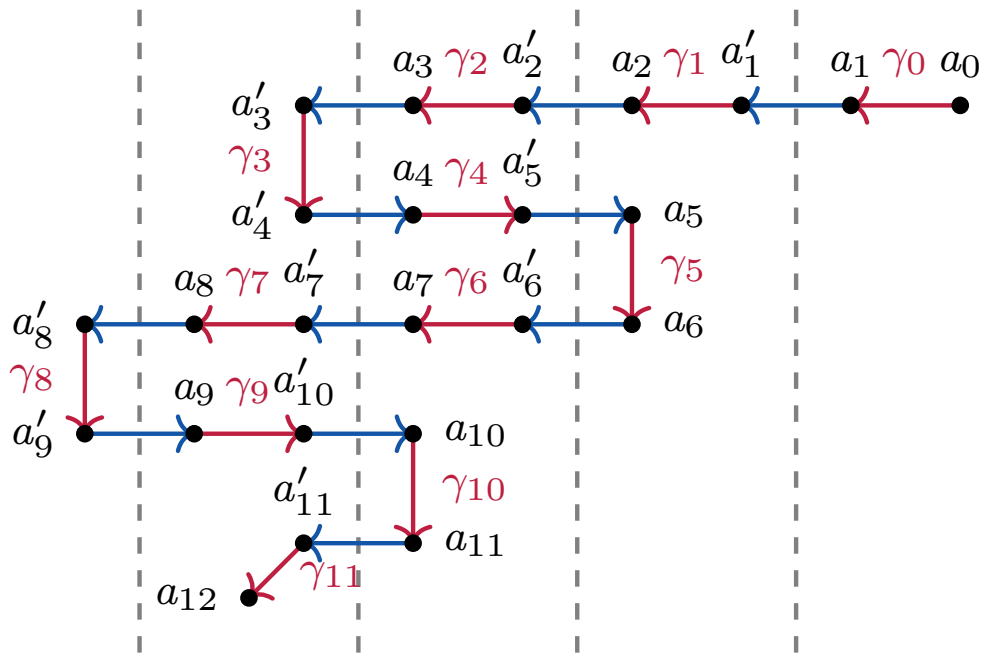
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- a_{n+1} is not in the closure of previous elements.



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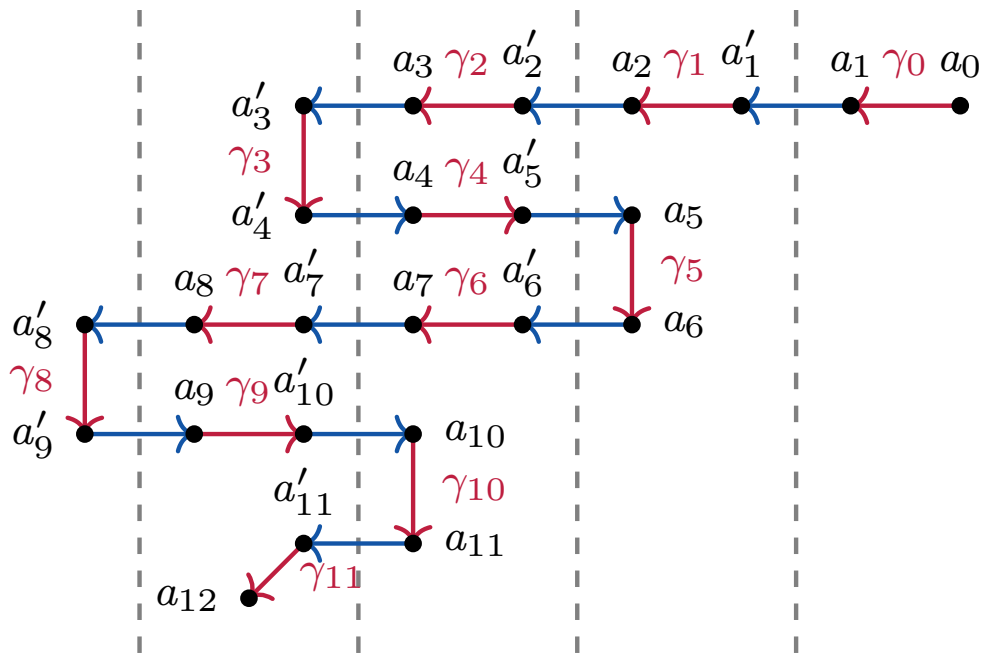
- a_{n+1} is in the closure of previous elements
  $\Rightarrow$ By modularity+IH,
  a_{n+1} is in closure of previous elements of its rank;
  we already know where it has to move
  (with IH guaranteeing independence of columns).
- a_{n+1} is not in the closure of previous elements.



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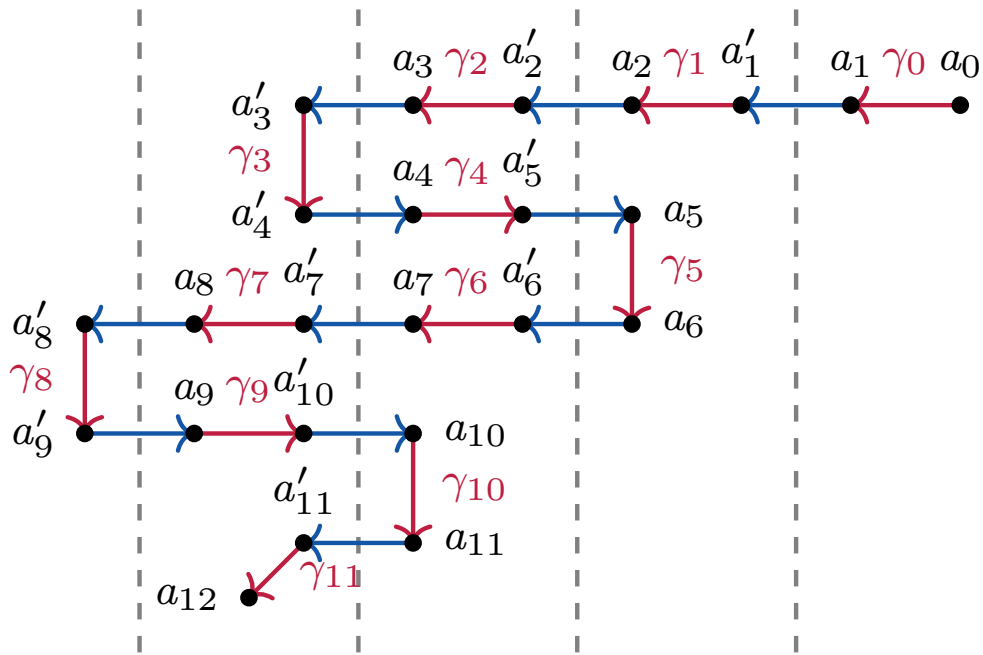
- a_{n+1} is in the closure of previous elements;
- a_{n+1} is not in the closure of previous elements
  $\Rightarrow$ By higher Neumann+pregeometry, move a_{n+1} and all of its translates to guarantee independence of columns.



Why these geometric conditions?

Induction splits into two cases:

- a_{n+1} is in the closure of previous elements;
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▸ Go back

The microsupported proof

Proof.

Let $w := \gamma_n \; x_{\iota_n}^{\epsilon_n} \; \gamma_{n-1} \; x_{\iota_{n-1}}^{\epsilon_{n-1}} \; \dots \; \gamma_1 \; x_{\iota_1}^{\epsilon_1} \; \gamma_0$.

$u := w$ after forgetting constants.

Let $\xi_k := \gamma_k \gamma_{k-1} \cdots \gamma_0$; $\Sigma := \{1, \xi_0, \dots, \xi_n\}$.

By microsupport: there is $V \subseteq \Omega$ open such that any two members of Σ either agree on V or have disjoint images.

$G_{\Omega \setminus V} \curvearrowright V$ is also microsupported. By lawlessness, there are $h_1, \dots, h_r \in G_{\Omega \setminus V}$ such that $u(h_1, \dots, h_r) \neq 1$.

Set $p_i := \prod_{j \in D} \xi_j h_i \xi_j^{-1}$ (D selects one index per disjoint image).

- For $a \in V$, $w(p_1, \dots, p_r)(a) = \xi_n u(h_1, \dots, h_r)(a)$.

By construction of V (and $1, \xi_n \in \Sigma$):

- (EITHER) $\xi_n(V) \cap V = \emptyset$: $w(p_1, \dots, p_n)(V) \cap V = \emptyset$;
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